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Education

Singapore University of Technology and Design, Bachelor of Engineering in Computer Science, 2026.

- **Cocurriculars**: Galois Group Mathematics Society, Google Developer Student Club (3DC).
- **Concentrations**: Minor in Artificial Intelligence, Specialisation in Visual Analytics and Computing.

Experiences

Creatz3D and Figure Factories, AI Engineer, 2025.

1. Productionised mesh thickening by training, testing, and deploying GraphSAGE models to thicken meshes.
2. Accelerated ML training pipelines by caching jobs, changing primitives, and vectorising functions.
3. Operationalised mesh generation by creating cron-scheduled Hunyuan3D-2 inference pipeline.
4. Supported ML inference pipelines by patching CUDA kernels for dependency compatibility.
5. Streamlined UGC production by designing human-in-the-loop generative AI orchestrations.

Proficiencies

- **Languages.** C, CSS, Dockerfile, HTML, Java, JavaScript, Makefile, Python, Shell, SnekScript, Swift, TeX.
- **Tools.** AWS, Azure, Cloudflare, Docker, FAISS, FastAPI, Figma, Firebase, Flask, Git(Hub), Gradio, Hugging Face, LangChain, llama.cpp, Matplotlib, MLX, Next.js, NumPy, Ollama, OpenRouter, Open WebUI, Paint.NET, PyTorch, Qdrant, React, SQLite, Shadcn, Streamlit, Tailwind, TensorFlow, Vercel, Whisper, n8n, pandas, scikit-learn.

Projects

PDPal, AI Legal Assistant, 2026.

1. Crafted backend architecture by defining targets, running experiments, and reasoning over trade-offs.
2. Tested and saw system outperform GPT-5.4 by 4.5% and Gemini 3.1 Pro by 9.1% on YPDYC oracle@3.
3. Improved frontend accessibility by designing responsive multilingual visual components.
4. Validated frontend accessibility with 73% of users fully recommending the product in preliminary studies.

SUTD404, SUTD Room Address Lookup Tool, 2026.

1. Trained Chinchilla-optimal Nanochat-style transformers to rewrite noisy SUTD room names.
2. Tested and saw model outperform SUTD Assistive Maps on test subset accuracy.
3. Optimised inference smoothness and speed by making it cached, debounced, and incremental.

Snekboros, "Vision World Models for Snake Playing", 2026.

1. Trained SIGReg-regularised convolutional GRU world models to play 1P Snake from 4x4 grid images.
2. Tested and saw model outperform random baseline by 89% on test set score.
3. Demonstrated the effect of planning horizon and sample count on survival.

Vibe Decoding, "An Unexpected Expected Next-Token Rule for Language Models", 2025.

1. Proposed aligning next-token embeddings with their expectation.
2. Tested Qwen 2.5 0.5B, Gemma 2 4B, and Llama 3.1 8B on BoolQ, NumerSense, PG-19, and WikiText-2.
3. Saw macro F1 and MRR of models scale positively with vibe strength on BoolQ and NumerSense.
4. Saw PPL of models negatively with vibe strength on PG-19 and WikiText-2.
5. Characterised asymptotic runtime of rule.
6. Determined interval of vibe strength at which rule becomes greedy.

BLIPS Regularisation, "Regularisation with Buffered Linear Interpolation by Positional Similarity", 2025.

1. Proposed using buffer to lerp between recent and current feature maps by positional similarity.
2. Made spectrogram images and resampled envelopes from STEAD subset by applying STFTs and Hilbert transforms.
3. Trained and tested CNNs, RNNs, and ViTs on train subset.
4. Tested and saw best ViT achieve 99.6% macro F1 on test subset.
5. Interpreted failures of best ViT by visualising its attention on false positives and negatives using Grad-CAM.

Recognition

- Showcased MLOps Projects, Singapore University of Technology and Design, 2026.
- Showcased Deep Learning Projects, Singapore University of Technology and Design, 2025.
- Information Systems Prize, Singapore University of Technology and Design & Singtel, 2024.
- Physical World Design 1D Award, Singapore University of Technology and Design, 2022.
- Malay (Special Programme) Top in Cohort Award, Temasek Secondary School, 2018.
- Mathematics and Science Book Prize, Temasek Secondary School, 2017.